

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
4	(a)	(i)	$\lambda = 0.20 \text{ [m]}$ (or by implication) (1) $v = \frac{0.20 \text{ [m]}}{0.040 \text{ [s]}}$ or $0.20 \text{ [m]} \times 25 \text{ [Hz]}$ or $5.0 \text{ [m s}^{-1}\text{]}$ or $d = 0.20 \text{ [m]} \times \frac{0.34 \text{ [s]}}{0.04 \text{ [s]}}$ (1) $d = 1.7 \text{ [m]}$ (1) Award 1 mark only if λ wrong but all else right		3		3	2	
		(ii)	Upside down, roughly same amplitude at least one complete cycle shown		1		1		
	(b)	(i)	[Point] that doesn't move or of no oscillation or of zero (minimum) amplitude or of destructive interference or equiv (1) 0.10 [m] ecf (1) Don't accept $\frac{\lambda}{2}$	2			2		
		(ii)	Points between [successive] nodes in phase or points a wavelength apart are in phase (1) Phase reversals at each node or equivalent (1)	2			2		
		(iii)	[Progressive] waves reflect [at clamp] (1) [Progressive] waves travelling in opposite directions [reflected covers this] and interfere or superpose (1) Accept specific case, e.g. waves travelling in opposite directions and interfere destructively at nodes	2			2		
		(iv)	Nodes closer together or shorter wavelength Accept more nodes or more antinodes		1		1		
			Question 4 total	6	5	0	11	2	0